



Academic Course Planning Software System at EEC Department

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ABSTRACT

The scheduling of student courses is an imperative as well as a trivial process that can cope with unnecessary industrial and graduation delays. The Department of Electrical and Communication (EECE) at the United Arab Emirates University (UAEU) is one such organization where students encounter problems conditional on many issues that may include improper course advising system, students' capability to pursue upright advice from advisers, knowledge, and adequate skills of the academic advisors, etc. Inappropriately advised students may endure issues such as lost time because of incorrect course selection, course planning with conflicting offering times, not selecting appropriate courses specific for offering semesters, wrongly selecting o discipline-specific electives, planning either too less or too many courses for a specific semester, etc. An Academic Course Planning Software System (ACPSS) is devised to help students in creating a course study plan to guide them select suitable courses at the registration time. The outcome of this course selection process, which takes the form of a detailed study plan, is recorded in a text file. The system is tested by more than 20 students, however, three test cases are included in this article.

CCS CONCEPTS

• Theory of computation; • Software and its engineering;

KEYWORDS

Course planning, Academic advising

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1 INTRODUCTION

Academic advisors and course planners are expected to exchange information needed to assist stakeholders in achieving their education and educational goals. This is a shared understanding and responsibility between a school trustee and the student. Advising is necessary for situations where an academic representative (faculty or staff) advises a university student about academic matters such as requirements needed for degree completion, course and career planning, and typical dialogues on how a course of study fits a particular academic or career interest [1, 2].

Course registration intended by a student comprises three main components: knowledge of the necessary and required pre-registration information, instructions, and guidelines for registering for courses, and lastly, guidelines on the next steps after the student has completed the course registration process. The process of registering for courses and what the students are required to do afterward is well established, however, the knowledge of what courses to register for is a noteworthy process and needs attention. A student is generally confused when selecting appropriate courses from an extensive pool of available courses.

Typical registration problems that are based on course planning and advising include students missing out on courses specific for alternating semesters, choosing exceedingly more or exceedingly fewer courses than expected as too few courses may increase in graduation duration period and too many may entail a heavy burden with loss of quality and course grades, advised or chosen courses with time conflicts or even greater time apart, for example, classes too early in the morning and too late in the afternoon.

The course advising resulting in study plans is moderately performed before course registration, however, students at the registration time encounter time conflicts which often results in losing other alternatives as classes fill up fast once registration starts, and it is also possible that a student may end up having all courses on two days of the week rather than evenly spread out on all four days of classes.

Students with such problems may suffer from delayed industrial training (one-semester-long training performed in an industry) and graduation duration due to unnecessary course selection or having missed out on important courses, dropping a complete semester because of a minimum number of course requirements (12 credits), or having a heavy semester load and may perform averagely due to a high number of courses, etc.

The ACPSS is developed to provide real-time plans for students and to enable them to maximize their opportunities in registering

for courses of their interest, as well as advise them to complete their degree requirements optimally. Frontal programming of the software is performed using Java and JavaScript computer programming languages.

The designed package assists and guides students in selecting appropriate courses for preparing a study plan. A typical course selection procedure begins with the student downloading a list of previously completed or in-class courses. The software then progresses through several procedures to advise students with an ideal list of the most appropriate courses. The choice, of course, is based on a field of knowledge built around all the courses of the electrical engineering program. The paper describes the complete operation of the course planning package that includes prioritized course selection, help menus, restrictions, etc. The software package has been tested to prepare 20 near-perfect study plans except for a minor modification to one of the plans to give an accuracy rate of 95%. The present document contains three examples of these plans.

2 THE LITERATURE REVIEW

In the early enrollment processes, institutional academic departments were responsible for enrolling students in courses. These preliminary procedures have been plagued by delays, a lack of resources, and management and administration problems. At a later stage, this perception of registration was examined from different perspectives, including course planning, course scheduling, course counseling, course registration, etc.

At the end of the 1990s, when the number of students enrolled increased, all the aforementioned issues also doubled. With this increase, the enrolment process has also had to be restructured to maximize the allocation of seats for this increased number of students enrolled. In addition, many institutions have started to provide more sophisticated, specialized, and multidisciplinary degree programs. Time disputes during the choice of courses for these programs have also increased. The primary purpose of course planning and recording systems is to select the most suitable courses without encountering temporal conflicts.

Computerized registration systems have sprung up and become popular over time. However, while there were major advantages to these systems, the main drawback was continued machine and network failures. It was anticipated and expected that most of the human errors, such as mistakes in inputting correct data would be detected by the system as they occurred, however, for errors that had not been foreseen, it was expected that some "off-line" data manipulation was necessary. Some of the first systems are touched upon in the following paragraphs.

The authors of this reference paper have proposed to design an online course selection system for students who are seeking to apply for a university degree and based on this design, propose improvements to a rule-based decision support system that enables them to choose a reasonably better university when selection is between a public and a private institution [3].

A student information system called MYGJU, developed by researchers in this reference paper can handle academic and financial information. This online registration is easy to use and prevents students from choosing incorrect and useless classes. However,

even after the launch of MYGJU, the system has not grown in popularity among students, and this is also discussed in this reference document [4].

The main authors of this study suggested, designed, and implemented several planning and advisory systems as early as 2005. Course planning and counseling algorithms were studied with parameters of a variable number of priority areas associated with each course and with different layouts of those domains. Most of these systems have been eliminated because of program changes and certain unexpected errors [5–9].

This background paper has examined many university systems to innovate an automated device for academic advice in a typical university system. A sense of developing and implementing a new model is discussed as a web application. A model is proposed to facilitate the tracking of student complaints and proposals with advisors and staff. The model is a computer-based system that helps school trustees provide quality, accurate and consistent advice to their students [10].

Genetic algorithms are used to design a course scheduling information strategy as the course schedule is considered a complex optimization problem and is difficult to handle manually. An optimal solution is sought by looking at potential solutions. The "Codeigniter" framework, reactive bootstrapping (display), and MySQL (database) are the main parameters of the genetic algorithm methodology. Course conflicts are significantly reduced with the help of this algorithm [11].

Several authors have produced a system of course planning assisted by artificial intelligence. Difficulties of the current scheduling system are also discussed in this paper and are compared with the new scheme. A navigator/server mode architecture is adopted to run the prototype of the course programming system [12].

The authors of this article have developed a class scheduling management system called "ClassSchedule" to address the problem of course planning and scheduling. The system can detect course conflicts, multi-user, and group course planning, and manage many resources together. A lazy loading feature is used for real-time operations [13].

Legaspi et al. have developed an online course planning system in which an avid algorithm is used for course planning and allocation to the faculty [14].

The taxonomy of important learning in software engineering courses is presented in this work, along with suggested curriculum goals and key elements including teaching and learning activities and feedback assessments. Evaluation, lessons learned, and the effect of the adjustment show that software engineering education has made great progress [15].

3 COURSE LEVEL PRIORITIES

There are approximately 2,400 students at the College of Engineering (COE). The COE is one of the nine colleges of the United Arab Emirates (UAE) University with an estimated total of 14,000 students. Electrical Engineering is one of the five departments with approximately 300 full-time students. The students complete 147 credit hours to fulfill their bachelor's degree requirements. Based on the quality of their grades and GPA (Grade Point Average), they typically take 4½ years with an average of 17 credits per semester,

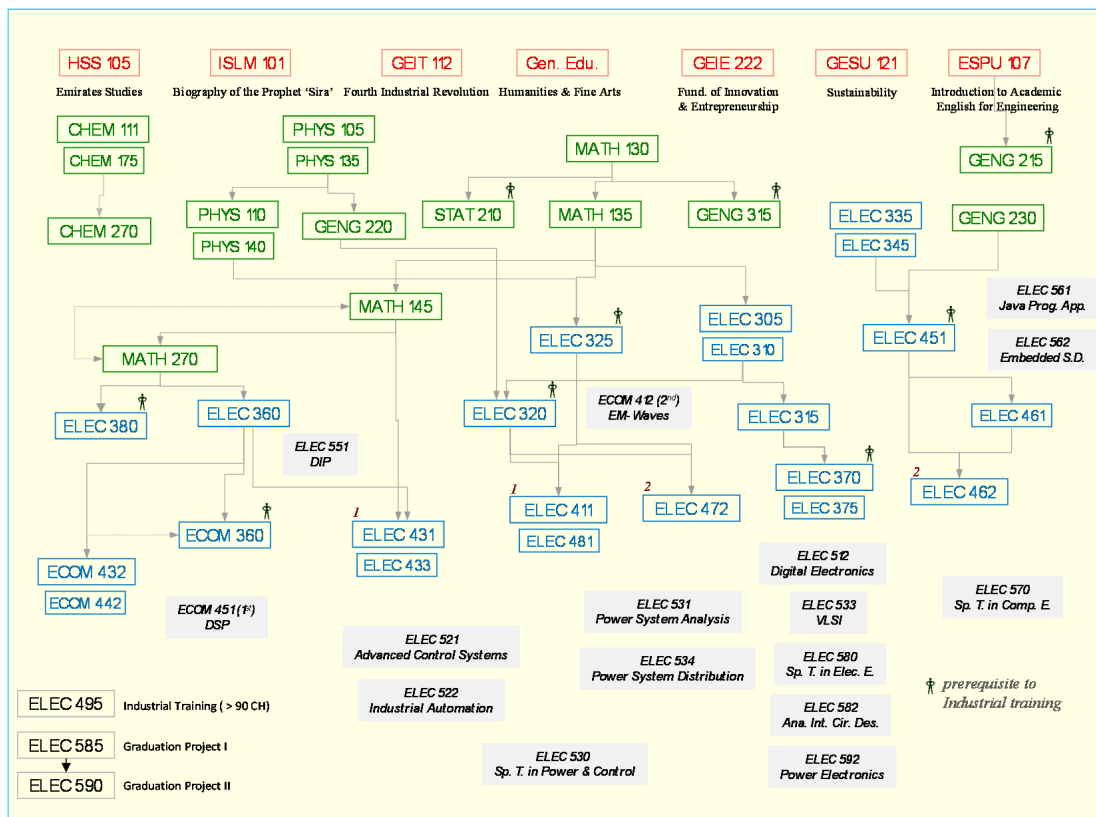


Figure 1: Electrical engineering curriculum course hierarchal chart

5 years with an average of 16 credits, 5½ years with an average of 15 credits, and 6 years with an average of 14 credits per semester to complete their degree requirements. Students with low educational attainment may take even longer because they sometimes fail classes.

Students take 52 courses divided into 6 main sections: General Education Requirements - 7 courses (21 credits), College Requirements - 15 courses (38 credits), Compulsory Specialization Requirements - 23 courses (55 credits) including 7 laboratory courses, Industrial Training - one course (15 credits), Graduation Projects - 2 courses (6 credits), and Elective Specialization Requirements - 4 courses (12 credits).

An organizational chart of courses, as shown in Figure 1, includes all mandatory and elective courses. The red background courses are General Education, the green color is College Requirements, the blue colors are required Specialist Requirements, and the colorless are the technical choices. The arrows indicate hierarchies. The asterisk on the course box shows that the specific course is required for industrial training. The numerical value of either '1' (1st semester) or '2' (2nd semester) specifies the course offering semesters.

Industrial Training (IT) is a complete semester (including Summer) course where a student spends the entire time with an industrial unit. IT eligibility begins after the completion of 94 or more credits and also depends on educational quality.

To establish the knowledge area of course planning, there are three priority levels for each course, as shown in Table 1. These priorities make it possible to define specific courses based on their importance in the selection of courses for the preparation of the curriculum. These are prioritized from highest to lowest and described as follows:

3.1 Course Level Priority I

The first level of priority is associated with industrial training (IT). This attribute is considered the highest priority for the course selection procedure. Delaying a course that is required for IT may delay graduation. A numerical value of 1 is considered for this attribute for all courses that must be completed before IT. For example, ELEC 451 is required for IT whereas ELEC 462 is not needed and therefore is assigned a '0' value in the first appended level. All specialization electives and almost all General Education courses are not required for industrial training. Whereas all College Requirement courses and most of the core courses are required.

3.2 Course Level Priority II

It is associated with the total number of courses that are linked to a course or that are open for a particular course. A numerical value of one is considered for each course that is opened by a certain course. For example, ELEC 335 is responsible to open five different courses in all associated hierarchical levels which are ELEC 451, ELEC 461,

Table 1: Course knowledge area with priority levels

	Course Code	CH	I	II	III		Course Code	CH	I	II	III
1	HSS_105	3	0	0	0	33	ELEC_370	3	1	5	1
2	ISLM_101	3	0	0	0	34	ELEC_375	1	1	5	1
3	GEIE_222	3	0	0	0	35	ELEC_380	3	1	0	0
4	GEIT_112	3	0	0	0	36	ELEC_431	3	0	3	1
5	GESU_121	3	0	0	0	37	ELEC_433	1	0	3	1
6	GEDU_000	3	0	0	0	38	ELEC_411	3	0	0	0
7	ESPU_107	3	1	1	1	39	ELEC_481	1	0	0	0
8	STAT_210	3	1	0	0	40	ECOM_432	3	0	0	0
9	GENG_215	2	1	0	0	41	ECOM_442	1	0	0	0
10	GENG_220	3	1	7	3	42	ELEC_451	3	1	4	2
11	MATH_130	3	1	35	5	43	ELEC_461	1	1	3	1
12	MATH_135	3	1	32	4	44	ELEC_462	3	0	0	0
13	MATH_145	3	1	13	3	45	ELEC_472	3	0	3	1
14	MATH_270	3	1	12	3	46	ELEC_495	15	1	2	2
15	PHYS_105	3	1	12	4	47	ELEC_585	3	1	1	1
16	PHYS_135	1	1	12	4	48	ELEC_590	3	1	0	0
17	PHYS_110	3	1	8	3	49	ECOM_451	3	0	0	0
18	PHYS_140	1	1	8	3	50	ELEC_512	3	0	0	0
19	CHEM_111	3	1	1	1	51	ELEC_551	3	0	0	0
20	CHEM_175	1	1	1	1	52	ELEC_561	3	0	0	0
21	GENG_315	3	1	0	0	53	ELEC_562	3	0	0	0
22	CHEM_270	3	1	0	0	54	ELEC_533	3	0	0	0
23	GENG_230	3	1	5	2	55	ECOM_412	3	0	0	0
24	ELEC_305	3	1	15	3	56	ECOM_570	3	0	0	0
25	ELEC_310	1	1	15	3	57	ELEC_521	3	0	0	0
26	ELEC_315	3	1	7	2	58	ELEC_522	3	0	0	0
27	ELEC_320	3	1	5	2	59	ELEC_592	3	0	0	0
28	ELEC_325	3	1	5	2	60	ELEC_531	3	0	0	0
29	ELEC_335	3	1	5	2	61	ELEC_534	3	0	0	0
30	ELEC_345	1	1	5	2	62	ELEC_580	3	0	0	0
31	ELEC_360	3	1	10	2	63	ELEC_582	3	0	0	0
32	ECOM_360	3	1	3	1	64	ELEC_530	3	0	0	0

ELEC 462, ELEC 562, and ELEC 570 as shown in Figure 1 as well as in Table 1. Related laboratory courses are considered to be part of their theoretical counterpart and thus are not included in the number of courses. However, specialization electives are included in this course count. In the course selection procedure, the course with a higher Level 2 value is prioritized over the other courses.

3.3 Course Level Priority III

This section addresses the course hierarchy levels associated with each course. For example, ELEC 305 has three forward hierarchal levels as shown in Figure 1, also indicated in Table 1. These different hierarchies are shown as ELEC 305 -> ELEC 315 -> ELEC 370 with five associated electives, ELEC 305 -> ELEC 320 -> ELEC 472 with three associated electives of (ELEC 530, ELEC 531, & ELEC 534), and ELEC 305 -> ELEC 320 -> ELEC 411. Students who miss courses from long hierarchical levels can delay their industrial training, including graduation.

4 THE ACPSS PACKAGE

The procedures of the ACPSS software begin with a typical student selecting all courses completed during the semester. The course selection is saved in a database that includes the student's name, university ID, email address (optional), year of study-plan preparation (optional), and mobile number (optional). Three sets of files are initially created in the database. An initial file listing all courses in the curriculum included the course number, course name, credit hours, and courses associated with three priority levels. A second file lists all courses offered during the fall semester alone, and a third file lists the courses offered during the spring semester.

The graphical interface of the developed software, as shown in Figure 2, contains several selectable buttons and text boxes. The set of buttons includes Save, Select Semester, Passed Courses, Courses, Chart, Reset, and Exit. The Passed Course button selects all passed and the current semester taken courses. The Save button displays this selection in the right viewport along with the selected list in a file. The Select Semester button chooses the specific starting semester and year of the study plan. The starting semester/year for the

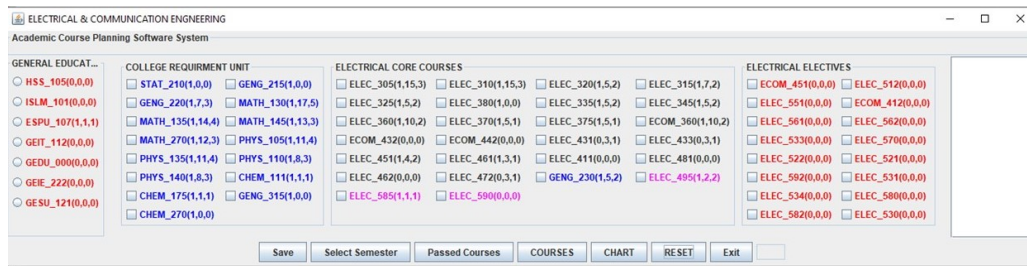


Figure 2: Graphical User Interface of the ACPSS package

three test cases is Spring 2023. The Courses button displays Table 1. The Chart button displays the hierarchical chart of Figure 1. The Reset button resets the course selection procedure whereas the Exit button exits the software package. In addition, selecting a course (not the course checkbox) displays all the necessary information about the course. Figure 3 shows examples of three typical courses.

The package starts with the student ID and Name query. The student then selects all successful courses in the course of the semester, then the plan for the beginning of the semester for the creation of the study plan. The selected courses disappear from the main window once the semester for selecting the specific course has finished. In addition to this, only a specific set of courses are displayed dependent on prerequisite requirements. This recommendation is based on the display of only the required courses for which the student has completed the prerequisites. With this recommendation, only courses eligible for a specific semester are presented to the student to avoid choosing courses not proposed for a specific semester. Figure 4 shows one such selection of the interface.

There is a small text field beside the Exit window that displays the total credited hours of the selected classes. A course selection error may be corrected by deselecting the course button and course credits are removed from this field. However, if an error is noted after the "Save" button, the entire process can be restarted using the "RESET" button. Three examples of test designs are presented in Tables 2, 3, and 4, respectively.

4.1 Test Case 1

The list of all courses selected (semester format) after the courses currently followed and passed 29 credits is presented in Table 2. The creation of the study plan begins with the Spring 2023 semester and continues until the Fall 2026 semester. The course selection is based on the designed three priority levels. The first stage confirms that the course is required for industrial training and delaying such a course may delay the completion of studies. The courses specific to industrial training are indicated by a human-like figure appended on the right upper corner, of the course boxes as shown in the organization chart.

The second level priority is for opening several courses dependent on the selected one. A higher number associates the importance of the course to be taken earlier. Selection of such courses allows freedom of choice from more courses compared to a less number.

The third priority of course hierarchy is next considered. A higher number associated with a course shows several forward

hierarchical levels linked to the specific course as indicated in Table 1 and Figure 1, respectively.

For Test Case 1, this typical student started the degree program in Spring 2022. The student comes with a low CGPA and is only allowed to plan 16 credits per semester. Therefore, suitable credit hours selection for each subsequent semester plans to graduate in eight more semesters.

4.2 Test Case 2

This typical student has already completed 53 credit hours of coursework. The student is of good academic standing and is therefore allowed to take more than 16 credits per semester. The student is planning to complete the degree requirements within 6 semesters starting from Spring 2023. It can also be seen that in almost all semesters, compulsory courses are supplemented with General Education Requirement courses, and this is for the reason to balance the difficulty level of courses each semester.

The student adopts to take heavier course load in the early semesters so that he/she can concentrate more on the Capstone graduation projects courses of ELEC 585 and ELEC 590, respectively. The 'Gen. Edu.' (Humanities and Fine Arts Group) course selected in Spring 2025 is the only course where the students have a choice to select one out of four offered courses.

4.3 Test Case 3

This student has already completed 39 credit hours of coursework. Although, this typical student has an average GPA, however, is occasionally allowed to take more than 16 credits. This student takes one more semester to graduate compared to the Test Case 2 example. Again, General Education courses supplement the difficulty level of core courses in a specific semester. However, Spring and Fall 2023 semesters are more concentrated on College and Compulsory Specialization Requirements and the student needs to focus more on these two semesters.

5 CONCLUSION

Student course planning is a necessary development to enable students to undertake academic requirements effortlessly. This paper presented an algorithm that helps students to create a complete study plan with the most suitable courses offered in specific semesters. An Academic Course Planning Software System is devised that is constructed around three-course level priorities to direct students to select from five to seven courses. There are 20 study plans generated using the ACPSS package. Only one of the

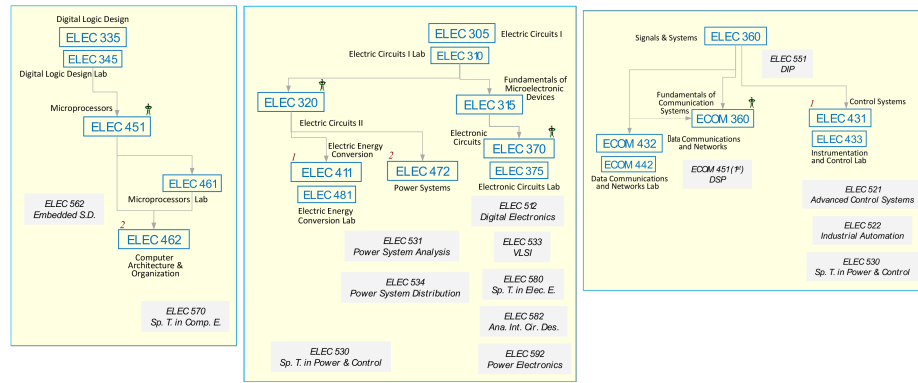


Figure 3: Course-level hierarchal examples of ELEC 335, ELEC 305, and ELEC 360

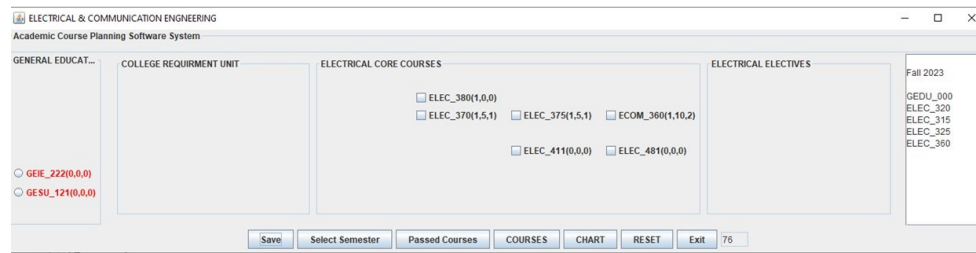


Figure 4: An example of a course selection procedure

Table 2: A study plan for a student in Test Case 1

Student ID: Student Name:									
Passed Courses									
1	HSS 105	3	16	ELEC 305	3	29	GENG 215	2	Fall 2025
2	CHEM 111	3	17	ELEC 310	1	Total: 15/74			41 ELEC 495 15
3	CHEM 175	1	Total: 14/43						Total: 15/118
4	STAT 210	3	Fall 2023			Fall 2024			
5	PHYS 105	3	18	GENG 230	3	30	ELEC 380	3	Spring 2026
6	PHYS 135	1	19	ELEC 315	3	31	ELEC 461	1	42 ECOM 432 3
7	GENG 220	3	20	ELEC 325	3	32	ELEC 431	3	43 ECOM 442 1
8	ESPU 107	3	21	ELEC 335	3	33	ELEC 433	1	44 ELEC 534 3
9	MATH 130	3	22	ELEC 345	1	34	CHEM 270	3	45 ELEC 585 3
10	MATH 135	3	23	ELEC 360	3	35	GENG 315	3	46 Gen. Edu. 3
11	GEIE 222	3	Total: 16/59			Total: 14/88			Total: 13/131
Total: 29			Spring 2024			Spring 2025			Fall 2026
Spring 2023			24	ELEC 320	3	36	ELEC 462	3	47 ELEC 411 3
12	MATH 145	3	25	ECOM 360	3	37	ELEC 472	3	48 ELEC 481 1
13	MATH 270	3	26	ELEC 370	3	38	ELEC 562	3	49 ECOM 451 3
14	PHYS 110	3	27	ELEC 375	1	39	ISLM 101	3	50 ELEC 561 3
15	PHYS 140	1	28	ELEC 451	3	40	GEIT 112	3	51 ELEC 590 3
						Total: 15/103			52 GESU 121 3
									Total: 16/147

Table 3: A study plan for another student in Test Case 2

Student ID: Student Name:										
Passed Courses			18	ELEC 305	3	30	ELEC 375	1	Spring 2025	
1	ISLM 101	3	19	ELEC 310	1	31	ELEC 380	3	42	ELEC 522 3
2	GEIE 222	3	20	ELEC 335	3	32	CHEM 270	3	43	ELEC 562 3
3	GEIT 112	3	21	ELEC 345	1	33	GENG 315	3	44	ELEC 585 3
4	ESPU 107	3	Total: 53			Total: 16/86			45	GESU 121 3
5	STAT 210	3							46	Gen. Edu. 3
6	GENG 220	3	Spring 2023			Spring 2024			Total: 15/133	
7	MATH 130	3	22	ELEC 315	3	34	ELEC 461	1		
8	MATH 135	3	23	ELEC 325	3	35	ELEC 462	3	Fall 2025	
9	CHEM 111	3	24	ELEC 320	3	36	ELEC 472	3	47	ELEC 411 3
10	CHEM 175	1	25	ELEC 360	3	37	ECOM 432	3	48	ELEC 481 1
11	PHYS 105	3	26	ELEC 451	3	38	ECOM 442	1	49	ELEC 431 3
12	PHYS 135	1	27	GENG 215	2	39	ELEC 533	3	50	ELEC 433 1
13	PHYS 110	3	Total: 17/70			40	HSS 105	3	51	ELEC 561 3
14	PHYS 140	1				Total: 17/103			52	ELEC 590 3
15	MATH 145	3	Fall 2023						Total: 14/147	
16	MATH 270	3	28	ECOM 360	3	Fall 2024				
17	GENG 230	3	29	ELEC 370	3	41	ELEC 495	15		
						Total: 15/118				

Table 4: A study plan for another student in Test Case 3

Student ID: Student Name:											
PC			Spring 2023			Spring 2024			Spring 2025		
1	GEIT 112	3	16	MATH 270	3	29	ELEC 461	1	41	ELEC 495	15
2	GESU 121	3	17	GENG 230	3	30	ELEC 462	3	Total: 15/116		
3	Gen. Edu.	3	18	ELEC 305	3	31	ECOM 360	3			
4	ESPU 107	3	19	ELEC 310	1	32	ECOM 432	3	Fall 2025		
5	STAT 210	3	20	ELEC 335	3	33	ECOM 442	1	42	ELEC 411	3
6	GENG 220	3	21	ELEC 345	1	34	ELEC 380	3	43	ELEC 481	1
7	MATH 130	3	22	ELEC 325	3	Total: 14/87			44	ECOM 451	3
8	MATH 135	3	Total: 17/56						45	ELEC 551	3
9	CHEM 111	3				Fall 2024			46	ELEC 585	3
10	CHEM 175	1	Fall 2023			35	ELEC 370	3	47	ISLM 101	3
11	PHYS 105	3	23	ELEC 315	3	36	ELEC 375	1	Total: 16/132		
12	PHYS 135	1	24	ELEC 320	3	37	ELEC 431	3			
13	PHYS 110	3	25	ELEC 360	3	38	ELEC 433	1	Spring 2026		
14	PHYS 140	1	26	ELEC 451	3	39	GENG 315	3	48	ELEC 472	3
15	MATH 145	3	27	CHEM 270	3	40	HSS 105	3	49	ELEC 522	3
Total: 39			28	GENG 215	2	Total: 14/101			50	ELEC 580	3
						Total: 17/73					

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